

Assignment 5

1. $E(3X^2 - 2X + 5) = ?$

X	:	-2	-1	0	1	2
$P(X=x)$	:	0.2	0.3	0.15	0.25	0.1

$$E(X) = \sum p_i x_i = -0.25, \quad E(X^2) = \sum p_i x_i^2 = 1.75$$

$$\begin{aligned} E(3X^2 - 2X + 5) &= 3E(X^2) - 2E(X) + 5 \\ &= 3(1.75) - 2(-0.25) + 5 \\ &= \boxed{10.75} \end{aligned}$$

2. $E(X^2) = \sum p_i x_i = \boxed{1.75}$

3. $X : 1, 2, 3, \dots, \infty$

$$P(X=x) = \frac{1}{2^x}$$

$$P(Y=1) = P(X \text{ is even})$$

$$= P(X=2) + P(X=4) + P(X=6) + \dots$$

$$= \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \dots$$

$$= \frac{1}{4} \left(1 + \frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots \right)$$

$$= \frac{1/4}{1 - 1/4} = \frac{1}{3}$$

$$\boxed{P(Y=1) = \frac{1}{3}}, \quad P(Y \neq 0) = 1 - P(Y=1) = \boxed{\frac{2}{3}}$$

$$4. P(X=x) = \frac{x}{15} ; x = 1, 2, 3, 4, 5$$

$$E(X), E(2X^2+1), E(3X+8)$$

$$E(X) = \sum p_i x_i = 1\left(\frac{1}{15}\right) + 2\left(\frac{2}{15}\right) + 3\left(\frac{3}{15}\right) + 4\left(\frac{4}{15}\right) + 5\left(\frac{5}{15}\right)$$

$$E(X) = \frac{1}{15} + \frac{4}{15} + \frac{9}{15} + \frac{16}{15} + \frac{25}{15}$$

$$E(X) = \frac{11}{3} = \boxed{3.67}$$

$$E(2X^2+1) = 2E(X^2) + 1$$

$$= 2 \sum p_i x_i^2 + 1$$

$$= 2\left(\frac{15}{3}\right) + 1$$

$$= \boxed{31}$$

$$E(3X+8) = 3 \cdot E(X) + 8$$

$$= 3\left(\frac{11}{3}\right) + 8 = \boxed{19}$$

$$5. E(X) = 10, V(X) = 25$$

$$Y = aX - b, E(Y) = 0, V(Y) = 1$$

$$V(aX - b) = a^2 V(X) = 1$$

$$a^2 = \frac{1}{25} \rightarrow a = \pm \frac{1}{5} \rightarrow \text{As its positive}$$

$$E(aX - b) = a E(X) - b = \frac{1}{5}(10) - b = 0 \rightarrow b = 2$$

$$\boxed{a = \frac{1}{5}, b = 2}$$

6. $E(X) = 2$, $V(X) = 5$

$$E(3X+2) = 3E(X) + 2 = 3(2) + 2 = 6 + 2 = 8$$

$$V(3X+2) = 3^2 \cdot V(X) = 9(5) = 45$$

7. $V(X+2) = V(X)$

$$V(3X+2) = 9V(X)$$

8. $E(X_1) = 5$, $E(X_2) = -2$, $E(X_3) =$
 $V(X_1) = 5$, $V(X_2) = 3$, $V(X_3) =$

(i) $E(X_1 + X_2) = E(X_1) + E(X_2) = 3$
 $V(X_1 + X_2) = V(X_1) + V(X_2) = 8$

(ii) $E(X_1 - X_2) = E(X_1) - E(X_2) = 7$
 $V(X_1 - X_2) = V(X_1) + V(X_2) = 8$

(iii) $E(2X_1 + 3X_2 - 5) = 2E(X_1) + 3E(X_2) - 5 = -1$
 $V(2X_1 + 3X_2 - 5) = 4V(X_1) + 9V(X_2) = 47$

9.	X	:	1	2	3	4	5	6
	P(X=x)	:	1/6	1/6	1/6	1/6	1/6	1/6

$$E(X) = \sum p_i x_i = \frac{7}{2}$$

(i) $E(S) = n \binom{7}{2} = \boxed{\frac{7n}{2}}$

(ii) $E(P) = E(X_1) \cdot E(X_2) \cdot E(X_3) \cdots E(X_n)$
 $= [E(X)]^n = \boxed{\left(\frac{7}{2}\right)^n}$

$y \backslash x$	0	1	2	Total
1	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{3}{8}$
2	0	$\frac{2}{8}$	$\frac{4}{8}$	$\frac{6}{8}$
3	0	$\frac{3}{8}$	$\frac{6}{8}$	$\frac{9}{8}$
Total	0	$\frac{6}{8}$	$\frac{12}{8}$	

(i) $E(X)$

$$10. f(x, y) = \begin{cases} \frac{xy}{8}, & 0 < x < 2, 1 < y < 3 \\ 0, & \text{otherwise} \end{cases}$$

$$(i) E(X) = \int_0^2 \int_1^3 x \cdot \left(\frac{xy}{8} \right) dy \cdot dx$$

$$= \int_0^2 \left[\frac{x^2 y^2}{16} \right]_1^3 dx \rightarrow \int_0^2 \frac{x^2}{8} \cdot \frac{9-1}{2} dx = \int_0^2 \frac{x^2}{8} \cdot 4 dx = \int_0^2 \frac{x^2}{2} dx$$

$$= 4 \left[\frac{x^3}{24} \right]_0^2 = 4 \left[\frac{8}{24} \right] = \frac{4}{3}$$

$$(ii) E(Y) = \int_1^3 \int_0^2 y \cdot \frac{xy}{8} dx dy$$

$$= \int_1^3 y \cdot \left[\frac{x^2}{24} \right]_0^2 dy \rightarrow \int_1^3 y \cdot \frac{8}{24} dy$$

$$= \frac{8}{24} \left[\frac{y^2}{2} \right]_1^3 = \frac{8}{24} \left[\frac{9}{2} - \frac{1}{2} \right] =$$

$$(ii) E(Y) = \int_0^3 \int_0^2 \frac{xy^2}{8} dx \cdot dy$$

$$= \int_0^3 \left[\frac{y^2}{8} \left[\frac{x^2}{2} \right]_0^2 dy \rightarrow \int_0^3 \frac{y^2}{8} \left[\frac{4}{2} \right] dy$$

$$= \frac{1}{4} \left[\frac{y^3}{3} \right]_0^3 = \frac{1}{4} \left[\frac{27}{3} - \frac{1}{3} \right] = \frac{1}{4} \left[\frac{26}{3} \right]$$

$$= \boxed{\frac{13}{6}}$$

$$(iii) E(2X+3Y) = 2E(X) + 3E(Y) = 2\left(\frac{4}{3}\right) + 3\left(\frac{13}{6}\right)$$

$$= \frac{8}{3} + \frac{13}{2} = \boxed{\frac{55}{6}}$$

11.	X :	1	2	3	4	5	6
	P(X=x) :	1/6	1/6	1/6	1/6	1/6	1/6

$$(i) E(X) = \sum p_i x_i = \boxed{\frac{7}{2}}$$

(ii)	Y :	1	2	3	4	5	6
	P(Y=y) :	1/6	1/6	1/6	1/6	1/6	1/6

$$E(Y) = \frac{7}{2}$$

$$E(X-Y) = E(X) - E(Y) = \boxed{0}$$

$$(iii) E(2X-3Y) = 2E(X) - 3E(Y)$$

$$= 7 - \frac{21}{2}$$

$$= \boxed{-\frac{7}{2}}$$

$$12. \mu_n' = n! = E(X^n)$$

$$\mu_1' = 1, \mu_2' = 2!, \mu_3' = 3!, \dots$$

$$M_0(t) = E(e^{tx})$$

$$M_0(t) = 1 + t + t^2 + t^3 + \dots$$

$$M_0(t) = \frac{1}{1-t}$$

$$13. f(x) = \begin{cases} ke^{-kx}, & x > 0, k > 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$M_0(t) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx$$

$$= k \int_0^{\infty} e^{(t-k)x} dx = k \left[\frac{e^{(t-k)x}}{t-k} \right]_0^{\infty}$$

$$= k \left[\infty - \frac{1}{t-k} \right] = \frac{k}{k-t}$$

$$\mu_1' = \frac{d}{dt} [M_0(t)]_{t=0} = \frac{d}{dt} \left(\frac{k}{k-t} \right)$$

$$\mu_1' = \frac{(k-t) \cdot (0) - k(-1)}{(k-t)^2} = \frac{k}{(k-t)^2} = \frac{1}{k}$$

$$\mu_2' = \frac{(k-t)^2 \cdot (0) - k \cdot 2(k-t) \cdot (-1)}{(k-t)^4} = \frac{2k^2 + 2k(k-t)}{(k-t)^4}$$

$$= \frac{2k(k-t)}{(k-t)^4} = \frac{2k}{(k-t)^3} = \frac{2k^2 + 2k^2 - 2kt}{(k-t)^4} = \frac{2}{k^2}$$

$$V(X) = \mu_2' - (\mu_1')^2 = \boxed{\frac{1}{k^2}}$$

$$14. f(x) = \begin{cases} 1/k, & 0 < x < k \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} M_0(t) &= \frac{1}{k} \int_0^k e^{tx} dx = \frac{1}{k} \left[\frac{e^{tx}}{t} \right]_0^k \\ &= \frac{1}{kt} \left[e^{tk} - 1 \right] = \frac{1}{kt} (e^{tk} - 1) \end{aligned}$$

(1) n^{th} moment about origin

$$\begin{aligned} \mu_n' &= \int_0^k x^n \cdot \frac{1}{k} dx = \frac{1}{k} \left[\frac{x^{n+1}}{n+1} \right]_0^k \\ &= \frac{1}{k} \left[\frac{k^{n+1}}{n+1} \right] = \boxed{\frac{k^n}{n+1}} \end{aligned}$$

$$(2) \text{ Mean} = \mu_1' = \frac{k^1}{1+1} = \boxed{\frac{k}{2}}$$

$$\begin{aligned} (3) \text{ Variance } (\sigma^2) &= E[X^2] = \mu_2' = \frac{k^2}{3} \\ &= E[X^2] - [E[X]]^2 = \boxed{\frac{k^2}{12}} \end{aligned}$$

$$15. f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$M_0(t) = \int_0^1 e^{tx} \cdot 1 dx$$

$$= \left[\frac{e^{tx}}{t} \right]_0^1 = \left[\frac{e^t - 1}{t} \right]$$

(1) x^{th} moment (μ'_x)

$$\mu'_x = \int_0^1 x^x \cdot 1 dx = \left[\frac{x^{x+1}}{x+1} \right]_0^1 = \boxed{\frac{1}{x+1}}$$

(2) Mean (μ'_1) = $\int_0^1 x \cdot 1 dx = \left[\frac{x^2}{2} \right]_0^1 = \boxed{\frac{1}{2}}$

(3) $\text{Var}(\sigma^2) = E[x^2] - [E[x]]^2$
 $= E[x^2] = \mu'_2 = \frac{1}{2+1} = \boxed{\frac{1}{3}}$

$$\text{Var}(x) = \mu'_2 - (\mu'_1)^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \boxed{\frac{1}{12}}$$

Ans.
12/02